

BACHELOR THESIS

Olivier Amacker

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Numerical Stability of Functional MRI Connectivity Biomarkers



Informatique et Systèmes de communication (ISC)

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University of Applied Sciences and Arts Western Switzerland

Faculty of Engineering and Architecture

Haute École d'Ingénierie de Sion

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Numerical Stability of Functional MRI Connectivity Biomarkers

Thesis submitted to the Bachelor degree programme
Informatique et Systèmes de communication (ISC) – Data engineering major
Haute École d'Ingénierie de Sion

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PROJECT REPOSITORY

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Abstract

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- The context and motivation for the research.
- The main objective or research question.
- A brief description of the methodology or approach used.
- The key results or findings.
- The main conclusion or implications of the work.

The abstract should be self-contained, clear, and usually does not exceed 250–300 words. It allows readers to quickly understand the purpose and outcomes of the thesis without reading the full document.

The abstract **must** be written in both French and English.

TODO Double check

As [Functional Magnetic Resonance Imaging \(fMRI\)](#) pipelines become more computationally heavy, requiring high-dimensional computations, it is crucial to assess how small numerical variability introduced by several different factors like the running [Operating System \(OS\)](#), parallelization strategies, and hardware architecture affects the results of these pipelines.

Building upon the paper “Numerical Variability of [Magnetic Resonance Imaging \(MRI\)](#) Graph Measures”^[1], this work goes a step further than looking at different [MRI](#) graph measures by also looking at [Major Depressive Disorder \(MDD\)](#) biomarkers, specifically the [Principal Component Analysis \(PCA\)](#) based biomarkers proposed by the paper “Extraction of robust functional connectivity patterns across psychiatric disorders using principal component analysis-based feature selection”^[2] by perturbing several steps of the pipeline. It also assesses the impact of numerically perturbed fMRIPrep preprocessing on FC Functional Connectivity (FC) matrices using the Numerical Population Variability Ratio (NPVR) metric.

To reproduce the results of the original study^[1], the results were obtained by running the fMRIPrep^[3] (25.2.5) preprocessing pipeline through a [custom Docker container](#) with floating-point arithmetic perturbations introduced by the [Verificarlo](#)^[4] and [Fuzzy](#)^[5] libraries. After preprocessing, [FC](#) matrices were obtained using Nilearn^[6] (0.13.1). Different absolute thresholds were applied to the [FC](#) matrices, and graph metrics were computed using NetworkX^[7] (3.6.1).

For the assessment of the [PCA](#) feature selection, two steps of the pipeline were perturbed: the correlation coefficient computation using a [specific Fuzzy Docker image](#), and the principal component analysis by forcing it to use 32-bit floating-point inputs rather than 64-bit.

The results given by the original work reproduction closely matched the original ones, except for a specific graph where the Y-axis values were hardcoded inside the original code. The [PCA](#) feature extraction appeared to be very stable under the introduced numerical perturbations, simulating [OS](#) level numerical variability. Giving the exact same results.

These findings suggest that the PCA features extraction is not only stable across different imaging sites and datasets, but also robust to numerical variability.

Keywords

medical data science, fMRI, fMRIPrep, numerical stability, benchmark, PCA, neuro imaging



Résumé

TODO Double check

Alors que les pipelines de [Imagerie par Résonance Magnétique Fonctionnelle \(IRMf\)](#) deviennent de plus en plus lourdes computationnellement, nécessitant des calculs en haute dimension, il est crucial d'évaluer comment de petites variations numériques introduites par plusieurs facteurs tels que le [Système d'Exploitation \(SE\)](#) utilisé, les stratégies de parallélisation et l'architecture matérielle affectent les résultats de ces dernières.

S'appuyant sur le papier « Numerical Variability of [IRMf](#) Graph Measures »^[1], ce travail va plus loin que l'examen de différentes mesures de graphes d'[Imagerie par Résonance Magnétique \(IRM\)](#) en examinant également les biomarqueurs du Trouble Dépressif Majeur (TDM), notamment les biomarqueurs basés sur l'[Analyse en Composantes Principales \(ACP\)](#) proposés par le papier "Extraction of robust functional connectivity patterns across psychiatric disorders using principal component analysis-based feature selection"^[2], en perturbant plusieurs étapes du pipeline. Il évalue également l'impact du prétraitement fMRIPrep numériquement perturbé sur les matrices de Connectivité Fonctionnelle (CF) en utilisant la métrique Numerical Population Variability Ratio (NPVR).

Pour reproduire les résultats de l'étude originale^[1], les résultats ont été obtenus en exécutant le pipeline de prétraitement fMRIPrep^[3] (25.2.5) dans un [conteneur Docker personnalisé](#) avec des perturbations arithmétiques en virgule flottante introduites par les bibliothèques [Verificarlo](#)^[4] et [Fuzzy](#)^[5]. Après prétraitement, les matrices de [CF](#) ont été obtenues à l'aide de Nilearn^[6] (0.13.1). Différents seuils absolus ont été appliqués aux matrices de [CF](#), et les métriques de graphes ont été calculées à l'aide de NetworkX^[7] (3.6.1).

Pour l'évaluation de la sélection de caractéristiques par [ACP](#), deux étapes du pipeline ont été perturbées : le calcul des coefficients de corrélation à l'aide d'une [image Docker Fuzzy spécifique](#), et l'analyse en composantes principales en la forçant à utiliser des entrées en virgule flottante 32 bits plutôt que 64 bits.

Les résultats de la reproduction de l'étude originale correspondent étroitement aux résultats originaux, à l'exception d'un graphe spécifique dont les valeurs de l'axe Y étaient codées en dur dans le code original. L'extraction de caractéristiques par [ACP](#) s'est avérée très stable sous les perturbations numériques introduites, simulant la variabilité numérique au niveau du [SE](#), tout en produisant des résultats strictement identiques.

Ces résultats suggèrent que l'extraction de caractéristiques par [ACP](#) est non seulement stable entre différents sites d'imagerie et jeux de données, mais également robuste face à la variabilité numérique.

Mots-clés

medical data science, fMRI, fMRIPrep, numerical stability, benchmark, PCA, neuro imaging

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- Dr Yohan Chatelain for his help on the Fuzzy perturbations
- The Advanced Telecommunications Research Institute International (ATR), for providing the necessary resources and environment to carry out this work.
- My family and friends back home, for their emotional support and encouragement during this journey.

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Writing a report is an exercise that involves both **content** and **form**. In this document, we aim to simplify the formatting aspect without making any assumptions about the content, specifically in the context of the ISC degree programme¹.

0.1 – The content of a thesis

The general structure of a bachelor thesis typically includes the following sections:

1. **Abstract:** A concise summary of the thesis, including the research question, methodology, results, and conclusions.
2. **Résumé:** A summary of the thesis in French.
3. **Acknowledgements:** [Optional] A section to thank those who supported your work.
4. **Table of Contents:** An organized list of chapters and sections.
5. **Introduction:** Presents the background/context, motivation, objectives, and scope and plan of the thesis.
6. **State of the Art / Literature Review:** Reviews existing research and situates the thesis within the academic context, if relevant to your work.
7. **Development and Methodology:** Describes the methods, materials, and procedures used in the research/thesis.
8. **Results:** Presents the findings of the research, often with tables, figures, and analysis.
9. **Discussion:** Interprets the results, discusses implications, and relates findings to the research question.
10. **Conclusion:** Summarizes the main findings and contributions, and suggests future work.
11. **References / Bibliography:** Lists all sources cited in the thesis.
12. **Appendices:** (Optional) Contains supplementary material such as raw data, code, or additional explanations.

¹ Here is how to add a footnote <https://isc.hevs.ch>

This structure may vary depending on the field of study, but these elements are commonly found in most bachelor theses. They are recommended for the *ISC Bachelor thesis* and should be adapted to the specific requirements of your thesis (e.g., if you have a state of the art section or not).

You can also change the order or the names of the sections, for instance, if you want to put the state of the art before the introduction, or if you want to add a section on methodology before the results.

0.2 – Academic titles

Please note that the academic titles of your supervisors and experts are important.

They should be included on the cover page, and you should use the correct title when addressing them in the acknowledgements section. For instance, a professor should be addressed as “Prof. [Name]”, while a doctor should be addressed as “Dr [Name]” (**without a colon!**). A professor who is also a doctor should be addressed as “Prof. Dr [Name]”.

If you are unsure about the title of your supervisor, co-supervisor, or expert, you can ask them directly or check their profile on the university website.

0.3 – Compiling the thesis

If you compile your thesis using the `typst` command line tool, or by using the `typst` extension in Visual Studio Code, please note that you must install the fonts used in this template. You can do so by running the following command in your terminal:

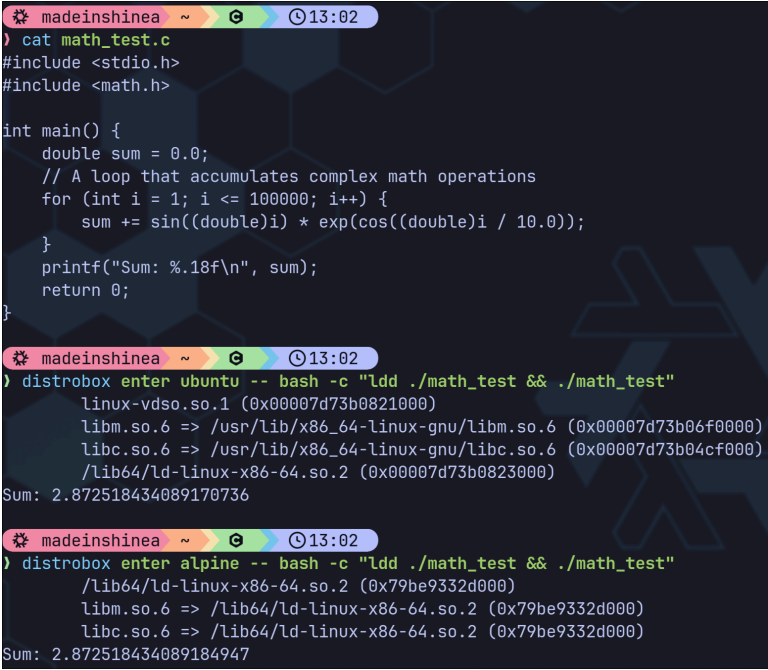
```
./fonts/install_fonts.sh
```

Chapter 1

Introduction

Functional Magnetic Resonance Imaging (fMRI) has revolutionized neuroscience by enabling non-invasive observation of brain activity through blood oxygenation level-dependent signals. A key application is the analysis of Functional Connectivity (FC), which examines temporal correlations between spatially distinct brain regions to construct connectivity matrices. These matrices serve as the foundation for deriving graph-theoretical metrics and biomarkers used in clinical research, particularly for conditions such as **Major Depressive Disorder (MDD)**.

As **fMRI** pipelines become increasingly computationally intensive, involving high-dimensional matrix operations and complex preprocessing steps, the question of numerical reliability becomes crucial. Small numerical perturbations introduced by factors such as **Operating System (OS)** differences (as you can see on Figure 1), hardware architecture, and parallelization strategies can propagate through the pipeline, potentially affecting downstream results.



```
madeinshinea ~ 13:02
) cat math_test.c
#include <stdio.h>
#include <math.h>

int main() {
    double sum = 0.0;
    // A loop that accumulates complex math operations
    for (int i = 1; i <= 100000; i++) {
        sum += sin((double)i) * exp(cos((double)i / 10.0));
    }
    printf("Sum: %.18f\n", sum);
    return 0;
}

madeinshinea ~ 13:02
) distrobox enter ubuntu -- bash -c "ldd ./math_test && ./math_test"
linux-vdso.so.1 (0x00007d73b0821000)
libm.so.6 => /usr/lib/x86_64-linux-gnu/libm.so.6 (0x00007d73b06f0000)
libc.so.6 => /usr/lib/x86_64-linux-gnu/libc.so.6 (0x00007d73b04cf000)
/lib64/ld-linux-x86-64.so.2 (0x00007d73b0823000)
Sum: 2.872518434089170736

madeinshinea ~ 13:02
) distrobox enter alpine -- bash -c "ldd ./math_test && ./math_test"
/lib64/ld-linux-x86-64.so.2 (0x79be9332d000)
libm.so.6 => /lib64/ld-linux-x86-64.so.2 (0x79be9332d000)
libc.so.6 => /lib64/ld-linux-x86-64.so.2 (0x79be9332d000)
Sum: 2.872518434089184947
```

Figure 1 - Example of the same program giving different results depending on the OS

In the context of multi-site studies and clinical applications, where reproducibility is essential, understanding and quantifying this numerical variability is critical. Unstable biomarkers could lead to unreliable clinical decisions or failed replication across research sites. However, systematic assessment of numerical stability in [fMRI](#) pipelines remains limited, particularly for advanced feature extraction methods.

The goal of this bachelor thesis, conducted in collaboration with the [Advanced Telecommunications Research Institute International \(ATR\)](#), is to investigate the numerical stability of [fMRI](#) connectivity biomarkers. This work was divided into three complementary parts.

First, the objective was to reproduce the results of Alizadeh et al., 2026^[1], which examined how numerical variability affects the sample size required for statistical significance and the stability of [FC](#) matrix graph metrics using the Numerical Population Variability Ratio (NPVR) metric.

Second, the focus shifted to the numerical stability of the [FC](#) matrices themselves, rather than their derived graph metrics, providing edge-wise stability analysis.

Finally, the numerical stability of a feature extraction method based on [Principal Component Analysis \(PCA\)](#) was assessed, building on the work of Yamashita et al., 2026^[2]. This method extracts [FC](#) biomarkers robust to different sites and datasets. The stability assessment involved perturbing the correlation coefficient computation (`np.corrcoef`) and forcing the [PCA](#) to use 32-bit floating-point inputs rather than 64-bit.

Chapter 2

State of the Art

In recent years, different tools have been created to help the assessment of numerical variability. Notably [Verificarlo](#) Denis et al., 2018 ^[4] which is a tool used to compile programs with [LLVM](#) level arithmetic perturbations and [Fuzzy](#) ^[5] which consists of already perturbed [Docker containers](#) for certain [Python](#) libraries like [Numpy](#) or [Scipy](#), and more recently [Fuzzy PyTorch](#) Gonzalez-Pepe et al., 2026 ^[8] which extends this approach to deep learning workflows.

Using these tools, the impact of numerical variability has been investigated across several neuroimaging domains such as functional, structural and diffusion imaging. As introduced in Section 1, Alizadeh et al., 2026 ^[1] evaluated the numerical variability of different graph measures derived from the [FC](#) matrices resulting from the widely used [fMRIPrep](#) Esteban et al., 2019 ^[3] preprocessing pipeline. Using the [NPVR](#) metric, they obtained values ranging from 0.1 to 0.2 for most graph metrics. These results were found to vary across brain regions, [FC](#) thresholds and confound regression strategies.

In structural imaging pipelines, Mirhakimi et al., 2025 ^[9] investigated numerical uncertainty in linear registration algorithms, demonstrating that small floating-point variations can lead to measurable differences in image alignment. Similarly, Chatelain et al., 2026 ^[10] quantified the impact of numerical variability on structural Magnetic Resonance Imaging (MRI) measures in Parkinson's disease, finding that numerical variation reached nearly one-third of population variability in multiple cortical and subcortical regions.

In diffusion imaging pipelines, Kiar et al., 2021 ^[11] showed that numerical uncertainty in analytical workflows leads to impactful variability in brain network measurements, revealing that different computational environments can produce substantially different connectivity results.

Interestingly, numerical variability has also been used as a resource rather than a limitation. Gonzalez-Pepe et al., 2026 ^[8] established that controlled numerical perturbations can serve as an effective data augmentation strategy for deep learning models, artificially expanding training datasets and improving model generalization.

However, these stability conclusions cannot be generalized across pipelines due to heterogeneous methodologies and implementations. While graph-theoretical metrics have been studied by Alizadeh et al., 2026 ^[1], edge-wise [FC](#) matrix stability remains unexplored. Additionally, the [PCA](#)-based feature extraction proposed by Yamashita et al., 2026 ^[2] has shown robustness across different sites and datasets, making its numerical stability verification crucial. This thesis addresses these gaps through edge-wise [FC](#) analysis and [PCA](#)-based extraction evaluation.

Chapter 3

Development and Methodology

This section describes the development process and methodology underlying this thesis. Beyond the thesis-specific code, this work included contributions to external repositories via pull requests, including fixes to the [ISC-HEI Typst thesis template](#), enhancements to [Yamashita et al.'s PCA-based feature extraction package](#), and a correction to the NPVR simulation in [Alizadeh et al.'s repository](#).

3.1 – Code availability

The code developed during this bachelor thesis is publicly available on [GitHub](#). The repository is organized using [git submodules](#) to separate concerns: the `bachelor-thesis-report` submodule contains this Typst report, while the `bachelor-thesis-project` submodule contains all analysis code, including preprocessing pipelines, analysis notebooks, and job submission scripts for the [ATR High Performance Computing \(HPC\)](#) cluster.

An interactive project website is available at olivier.amacker.dev/bachelor-thesis, providing browsable access to the four analysis notebooks developed with [Marimo](#) and their respective outputs, a daily journal documenting the project's progress, and the different presentations delivered during the thesis.

The development environment is managed via [Nix](#) and [uv](#), ensuring its reproducibility. The fuzzy fMRIPrep [Docker](#) container, built on [Verificarlo](#) and the Fuzzy libmath library, is published on [DockerHub](#) for reproducibility.

3.2 – Reproduction of Alizadeh et al., 2026

3.2.1 – NPVR Simulation

3.2.2 – Graph Metrics Stability Assessment

3.3 – Edge-wise FC Matrix Stability Analysis

3.4 – PCA-based Feature Extraction Stability

3.4.1 – Reproduction on SRPB and BMB Datasets

3.4.2 – FC Matrix Extraction Perturbation

3.4.3 – PCA Dimensionality Reduction Perturbation

Chapter 4

Results and Discussion

Chapter 5

Conclusion

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Appendix

Table of Acronyms

ACP :	Analyse en Composantes Principales
ATR :	Advanced Telecommunications Research Institute International
CF :	Connectivité Fonctionnelle
FC :	Functional Connectivity
HPC :	High Performance Computing
IRM :	Imagerie par Résonance Magnétique
IRMf :	Imagerie par Résonance Magnétique Fonctionnelle
MDD :	Major Depressive Disorder
MRI :	Magnetic Resonance Imaging
NPVR :	Numerical Population Variability Ratio
OS :	Operating System
PCA :	Principal Component Analysis
PR :	Pull Request
SE :	Système d'Exploitation
TDM :	Trouble Dépressif Majeur
fMRI :	Functional Magnetic Resonance Imaging

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Code Appendix

```

1 class Cons[+A](hd: A, tl: ⇒ MyStream[A]) extends MyStream[A] {
2   override def isEmpty: Boolean = false
3   override val head: A = hd
4   override lazy val tail: MyStream[A] = tl
5   override def #::[B >: A](elem: B): MyStream[B] =
6     new Cons[B](elem, this)
7
8   override def ++[B >: A](anotherStream: ⇒ MyStream[B]): MyStream[B] =
9     new Cons[B](head, tail ++ anotherStream)
10
11  override def foreach(f: A ⇒ Unit): Unit = {
12    f(head)
13    tail.foreach(f)
14  }
15 }

```

Listing 1 - Code included from the file example.scala

```

1 def mergeSort(arr):
2   if len(arr) ≤ 1:
3     return arr
4
5   mid = len(arr) // 2
6   leftHalf = arr[:mid]
7   rightHalf = arr[mid:]
8
9   sortedLeft = mergeSort(leftHalf)
10  sortedRight = mergeSort(rightHalf)
11
12  return merge(sortedLeft, sortedRight)
13
14 def merge(left, right):
15   result = []
16   i = j = 0
17
18   while i < len(left) and j < len(right):
19     if left[i] < right[j]:
20       result.append(left[i])
21       i += 1
22     else:
23       result.append(right[j])
24       j += 1
25
26   result.extend(left[i:])
27   result.extend(right[j:])
28
29   return result
30
31 mylist = [3, 7, 6, -10, 15, 23.5, 55, -13]
32 mysortedlist = mergeSort(mylist)
33 print("Sorted array:", mysortedlist)

```

Listing 2 - Second code included from the file example.scala

```

1 class Cons[+A](hd: A, tl: ⇒ MyStream[A]) extends MyStream[A] {
2   override def isEmpty: Boolean = false
3   override val head: A = hd
4   override lazy val tail: MyStream[A] = tl

```

```
5  override def #::[B >: A](elem: B): MyStream[B] =  
6      new Cons[B](elem, this)  
7  
8  override def ++[B >: A](anotherStream: ⇒ MyStream[B]): MyStream[B] =  
9      new Cons[B](head, tail ++ anotherStream)  
10  
11  override def foreach(f: A ⇒ Unit): Unit = {  
12      f(head)  
13      tail.foreach(f)  
14  }  
15 }
```

Listing 3 - Second code included from the file example.scala

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